

Suraj Rao

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Skills

Languages: C/C++, Python, SQL

Technical Stack: PyTorch, NumPy, Pandas, Matplotlib, Scikit-learn, OpenCV, CUDA, ONNX, TensorRT, Flask, Tkinter, Streamlit, FAISS, Git, Linux, LLM, Parallel Programming, Edge AI, Deep Learning, Computer Vision, Semantic Search

Work Experience

IIT Delhi, Delhi

Oct 2024 - Apr 2025

Research Associate

- Conducted research and experimentation to improve the detection accuracy for occluded vehicle scenarios in the Indian road context.
- Evaluated transformer-based detection models (DETR, DINO, Grounding DINO, Co-DETR) on the RSUD20K, IDD and BDD100K datasets to identify model strengths and areas for improvement.
- Enhanced the RSUD20K dataset to include high occlusion scenarios (>50%) using Grounding DINO pseudo-labeling and manual refinement. Developed a Python based GUI using Tkinter to efficiently correct false positives, significantly reducing manual annotation time.

Mantra Softech, Bangalore

Dec 2023 - June 2024

Research Associate Intern

- Conducted in-depth research to identify and deploy an optimal deep neural network for people counting and recognition, suitable for constrained edge deployment in video surveillance systems.
- Employed optimization techniques such as model quantization, pruning, multiprocessing, frame filtering, and batch inference for optimizing resource allocation and accommodating scalability for multiple cameras.
- Researched and evaluated open-source GenAI models, including DALL-E Mini, Stable Diffusion, CLIP, and various LLMs, to identify an effective solution for retrieving videos from a large database using text prompts.

Education

University of Maryland College Park

Aug 2025 - May 2027

Master of Science in Applied Machine Learning

GPA: 4.0/4.0

Relevant Coursework: Probability and Statistics, Optimization, Data Structures and Algorithms, Deep Learning, Computer Vision, Natural Language Processing, Cloud Computing

Indian Institute of Information Technology Pune

Aug 2020 - Aug 2024

Bachelor of Technology in Computer Science and Engineering

CGPA: 8.09/10

Relevant Coursework: Object Oriented Programming, Discrete Maths, Data Structures and Algorithms, Operating Systems, Computer Networks, Machine Learning, Deep Learning, Databases, Information Retrieval, Image Processing

Project Work

- Smart Crop Yield Prediction:** Developed a web application that uses Machine Learning models to predict the crop yield of over 60 crops. Training dataset (2007-2020) comprised parameters like location, soil type, temperature, humidity, nitrogen, phosphorus, potassium values in soil across India. Employed Random Forest Regression, Support Vector Regression, and XGBoost and observed the individual performances. Integrated XGBoost with Flask framework to build a web application that assists farmers by providing estimates of crop yield. *Python, Numpy, Pandas, Flask* [Github]
- CLIP-Based Multimodal Semantic Search Engine:** Built a semantic image search system for text-to-image and image-to-image retrieval across 5.4K+ animal images. Generated CLIP embeddings and indexed them using FAISS for fast similarity search. Created a modular web app with text and image query support. Delivered an interactive tool that returned top-5 semantically similar results in seconds. *Python, CLIP, FAISS, Flask* [Github]
- Human Face Detection:** Studied object detection and tracking techniques provided by the OpenCV library. Investigated deep learning alternatives for object detection, including Fast-RCNN, MobileNet-SSD, YOLO. Implemented a machine learning-based Haar Cascade Classifier to detect faces and demonstrated the algorithm's real-time performance using a webcam. *Python, OpenCV, Numpy* [Github]